Activity: Likelihood ratio tests with linear regression

Linear regression: comparing nested models

Suppose we observe iid data $(\mathbf{x}_1, Y_1), ..., (\mathbf{x}_n, Y_n)$ from the linear model

$$Y_i = \mathbf{x}_i^T \beta + \varepsilon_i$$

where $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, and $\mathbf{x}_i, \beta \in \mathbb{R}^p$. Suppose also that we partition $\beta = \begin{pmatrix} \beta_{(1)} \\ \beta_{(2)} \end{pmatrix}$, where $\beta_{(1)} \in \mathbb{R}^{q_1}$ and $\beta_{(2)} \in \mathbb{R}^{q_2}$, and $q_1 + q_2 = p$. Likewise partition $\mathbf{x}_i = \begin{pmatrix} \mathbf{x}_{i,(1)} \\ \mathbf{x}_{i,(2)} \end{pmatrix}$.

We are interested in testing the hypotheses

$$H_0 : \beta_{(2)} = 0 \qquad H_A : \beta_{(2)} \neq 0$$

This is equivalent to comparing the nested full and reduced models

Full model: $Y_i = \beta_0 + \beta_1 X_{i,1} + \cdots + \beta_{q_1 - 1} X_{i,q_1 - 1} + \beta_{q_1} X_{i,q_1} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i$

Reduced model: $Y_i = \beta_0 + \beta_1 X_{i,1} + \cdots + \beta_{q_1 - 1} X_{i,q_1 - 1} + \varepsilon_i$

Estimates

The unrestricted MLEs for β and σ^2 are:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \mathbf{x}_i^T \hat{\beta})^2$$

while the restricted MLEs (under H_0) are:

$$\hat{\beta}^0 = (\mathbf{X}_{(1)}^T \mathbf{X}_{(1)})^{-1} \mathbf{X}_{(1)}^T \mathbf{y} \qquad \hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \mathbf{x}_{i,(1)}^T \hat{\beta}^0)^2$$

Questions

1. Write down the likelihood $L(\beta, \sigma^2|\mathbf{X}, \mathbf{y})$ as a function of β and σ^2.

$$L(\beta, \sigma^2 | X, y) \propto \prod_{i=1}^{n} f(y_i | x_i, \beta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - x_i^T\beta)^2\right\}$$

$$= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i^T\beta)^2\right\}$$

2. Using question 1, show that

$$L(\hat{\beta}, \hat{\sigma}^2|\mathbf{X}, \mathbf{y}) = (2\pi\hat{\sigma}^2)^{-n/2} \exp\{-n/2\}$$

and

$$L(\hat{\beta}^0, \hat{\sigma}_0^2|\mathbf{X}, \mathbf{y}) = (2\pi\hat{\sigma}_0^2)^{-n/2} \exp\{-n/2\}$$

$$L(\hat{\beta}, \hat{\sigma}^2 | X, y) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\hat{\sigma}^2}\sum_i (y_i - x_i^T\hat{\beta})^2\right\}$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_i (y_i - x_i^T\hat{\beta})^2$$

$$\Rightarrow \frac{\sum_i (y_i - x_i^T\hat{\beta})^2}{\hat{\sigma}^2} = n$$

$$\Rightarrow L(\hat{\beta}, \hat{\sigma}^2 | X, y) = (2\pi\hat{\sigma}^2)^{-\frac{n}{2}} \exp\left\{-\frac{n}{2}\right\}$$

Likewise, $\hat{\sigma}_0^2 = \frac{1}{n}\sum_i (y_i - x_{i(1)}^T\hat{\beta}^0)^2$

$$\Rightarrow L(\hat{\beta}^0, \hat{\sigma}_0^2 | X, y) = (2\pi\hat{\sigma}_0^2)^{-\frac{n}{2}} \exp\left\{-\frac{n}{2}\right\}$$

3. The **likelihood ratio test** rejects when

$$\Lambda = \frac{\sup_{(\beta,\sigma^2)\in\Theta_0} L(\beta,\sigma^2|\mathbf{X},\mathbf{y})}{\sup_{(\beta,\sigma^2)\in\Theta} L(\beta,\sigma^2|\mathbf{X},\mathbf{y})} = \frac{L(\hat{\beta}^0,\hat{\sigma}_0^2|\mathbf{X},\mathbf{y})}{L(\hat{\beta},\hat{\sigma}^2|\mathbf{X},\mathbf{y})} \leq k$$

Using question 2, show that this is equivalent to rejecting when

$$\frac{SSE_{reduced}}{SSE_{full}} = \frac{\sum_{i=1}^{n}(Y_i - \mathbf{x}_{i,(1)}^T\hat{\beta}^0)^2}{\sum_{i=1}^{n}(Y_i - \mathbf{x}_i^T\hat{\beta})^2} \geq k'$$

for some k'.

$$\Lambda = \frac{(\hat{\sigma}_0^2)^{-\frac{n}{2}}}{(\hat{\sigma}^2)^{-\frac{n}{2}}} = \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_0^2}\right)^{\frac{n}{2}}$$

$$\Lambda \leq k \iff \frac{\hat{\sigma}^2}{\hat{\sigma}_0^2} \leq k^{\frac{2}{n}} \iff \frac{\hat{\sigma}_0^2}{\hat{\sigma}^2} \geq \underbrace{k^{-\frac{2}{n}}}_{k'}$$

and $$\frac{\hat{\sigma}_0^2}{\hat{\sigma}^2} = \frac{\frac{1}{n}\sum_i (Y_i - x_{i(1)}^T\hat{\beta}^0)^2}{\frac{1}{n}\sum_i (Y_i - x_i^T\hat{\beta})^2} = \frac{SSE_{reduced}}{SSE_{full}}$$

4. From question 3, conclude that the LRT is equivalent to rejecting when

$$\frac{(SSE_{reduced} - SSE_{full})/q_2}{SSE_{full}/(n-p)} \geq c$$

for some c.

$$\frac{SSE_{red}}{SSE_{full}} \geq k' \iff \frac{SSE_{red}}{SSE_{full}} - 1 \geq k' - 1$$

$$\iff \frac{SSE_{red} - SSE_{full}}{SSE_{full}} \geq k' - 1$$

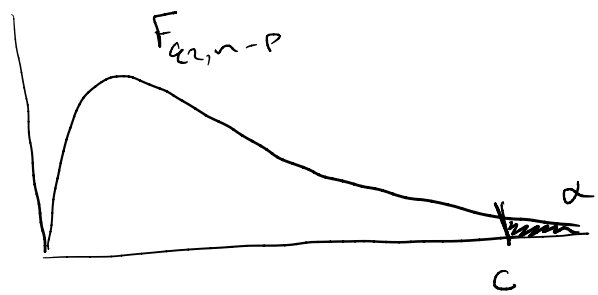
$$\iff \left(\frac{SSE_{red} - SSE_{full}}{SSE_{full}}\right) \cdot \frac{(n-p)}{q_2} \geq \underbrace{(k'-1)\cdot\frac{(n-p)}{q_2}}_{c}$$

$$\iff \frac{(SSE_{red} - SSE_{full})/q_2}{SSE_{full}/(n-p)} \geq c$$

5. It turns out that, under H_0, $\dfrac{(SSE_{reduced} - SSE_{full})/q_2}{SSE_{full}/(n-p)} \sim F_{q_2,n-p}$. Use this fact to choose a threshold c such that the test from question 4 has size α.

Want: $P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$

$$\Rightarrow P\Big(\underbrace{\frac{(SSE_{red} - SSE_{full})/q_2}{SSE_{full}/(n-p)}}_{\sim F_{q_2,n-p} \text{ under } H_0} \geq c \;\Big|\; \beta_{(2)} = 0 \Big) = \alpha$$

So, choose $c = F^{-1}_{q_2,n-p}(1-\alpha)$

i.e. upper α quantile of $F_{q_2,n-p}$ distribution